

Aakash Rajan

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EDUCATION

Indian Institute of Technology, Madras <i>Bachelor of Technology, Mechanical Engineering</i>	CGPA: 9.03 / 10 Aug 2024 – Present
Head Start Educational Academy <i>ISC, Class XII</i>	Percentage: 96% 2024
Head Start Educational Academy <i>ICSE, Class X</i>	Percentage: 89% 2022

SCHOLASTIC ACHIEVEMENTS

- Secured an All India Rank of **2130** in JEE Advanced 2024 - Top **0.25 percentile** out of 1M candidates
- National Winner**, Caterpillar Challenge 2026 – among **7,500+** competitors nationwide
- Awarded **Best Project in Batch** for ME2400 (Mechatronics) by the former HOD, Prof. Chandramouli

RESEARCH AND TECHNICAL EXPERIENCE

Research Intern – Video World Models <i>HiRo Lab, Indian Institute of Science (IISc) – Advisor: Prof. Ravi Prakash</i>	Jun 2026 – Present Bengaluru, India
- Working on making video world models more geometry-aware for closed-loop robot navigation; modifying, training, and running inference on models including the Gaussian World Model, DINOv3, and $\pi 0.5$. - Devised a memory-retrieval mechanism for WorldPack, improving SSIM by $\sim 3.5\%$ and LPIPS by $\sim 3\%$ over baseline on the LoopNav benchmark. - Prototyped RoPE-based trajectory packing and geometric/semantic attention conditioning (using DINOv3) on the Navigation World Model (NWM) backbone.	
Undergraduate Researcher – Multi-Agent RL for Air Traffic Control <i>Indian Institute of Technology Madras – Advisor: Prof. Rahul Meshram</i>	Ongoing Chennai, India
- Training cooperative multi-agent reinforcement learning policies for decentralized air-traffic control, as part of a challenge hosted by AI4REALNET and TU Delft. - Aiming to improve conflict resolution and cooperation among agents under dense, dynamic airspace constraints.	
Multi-Agent Autonomous Dump-Truck Simulator <i>Caterpillar National Tech Challenge</i>	Mar 2026 – Jun 2026 India
- Led a team of 3 to develop a Python-based multi-agent dump-truck simulator for autonomous dump packing, winning Caterpillar's national contest among 7,500+ competitors with a robust, deployable solution. - Solved the task selection and optimal assignment problem using Heuristics, the Hungarian Algorithm, and Monte Carlo Tree Search (MCTS). - Integrated vehicle dynamics and path planning (Hybrid A*-CBS) to optimize complex truck reverse maneuvers into dump targets.	
Hub Motor Cooling Jacket Design <i>Formula Student, Raftar Formula Racing</i>	Jul 2025 – Nov 2025 Chennai, India
- Designed a lattice-strut based cooling jacket for an electric hub motor, optimizing pressure drop and heat-transfer coefficient backed by real life testing of an LPBF manufactured prototype. - Developed a MATLAB iterative solver for cooling-loop head loss, coupled with multi-phase CFD thermal analysis to improve simulation accuracy.	
Automated Medicine Dispenser <i>Mechatronics Course Project (ME2400), IIT Madras – Advisor: Prof. Sushanta Panigrahi</i>	Apr 2026 – May 2026 Chennai, India
- Designed the end-to-end pipeline, selected and integrated sensors/actuators, built CAD models, assembled the system, and programmed its functionality. - Awarded Best Project in Batch by the former HOD, Prof. Chandramouli.	

TECHNICAL SKILLS

Languages: Python, Java, MATLAB

Frameworks & Robotics: PyTorch, ROS2, Reinforcement Learning, World Models, AI/ML Implementation

Softwares: CAD, FEA, CFD, Simulink

RELEVANT COURSEWORK

- **Robotics and Control:** Measurements, Instrumentation and Control | Basic Electrical Engineering
 - **Thermal and Fluids:** Applied Thermodynamics | Fluid Mechanics | Heat Transfer*
 - **Mechanics and Design:** Kinematics and Dynamics of Machinery | Machine Design* | Materials and Design
 - **Mathematics and Programming:** Linear Algebra | Differential Equations | Complex Networks* | Python Data Structures & Algorithms*
- * - Ongoing

POSITIONS OF RESPONSIBILITY

Aerodynamics and Cooling Engineer Apr 2025 – Dec 2025
Formula Student, Raftar Formula Racing *IIT Madras*

- * Part of the engineering team responsible for aerodynamic component design and manufacturing; gained hands-on knowledge of aerodynamics, cooling strategies, vehicle dynamics & power electronics integration.
- * Looked after precise procurement, manufacturing, and assembly of all carbon fiber components.

Music Contingent Head (Piano) Present
Inter-IIT Cultural Meet 8.0, IIT Madras *IIT Madras*

- * Led a team of 8 musicians as **Music Director** of Stagecoach, Drama Club's highest-budget annual event.
- * Represented IIT Madras at Inter-IIT Cultural Meet 8.0 as **Lead Keyboardist**, placing 4th among 23 IITs.
- * Completed all 8 grades by **ABRSM** with merit in performance piano; Performed in several stages across India.

Business Plan Presentation – Runners Up Jun 2025 – Sept 2025
Raftar Formula Racing, Formula Bharat 2026 *IIT Madras*

- * Part of the team that ideated and planned an entire business plan for **Formula Bharat 2026**, securing **2nd place** out of 40+ teams *across India*.
- * Worked on market analysis, **TCO analysis**, competitive analysis, **supply chain**, and CAC estimation; developed a pitch deck and finance sheets for the fictional company.